

NED INS system.

The navigation equation resolved in F^n is:

$$\underline{a}_{ib}^n = \underline{f}_{ib}^n + \underline{y}_{ib}^n = C_b^n \underline{f}_{ib}^b + \underline{y}_{ib}^n$$

We want to use this to get $\underline{\dot{v}}_{eb}^n$.

$$\underline{v}_{eb}^n = C_e^n \underline{v}_{eb}^e$$

$$\Rightarrow \underline{\dot{v}}_{eb}^n = \dot{C}_e^n \underline{v}_{eb}^e + C_e^n \underline{\dot{v}}_{eb}^e = \Omega_{en}^n C_e^n \underline{v}_{eb}^e + C_e^n \underline{\dot{v}}_{eb}^e$$

$$\text{Since } \underline{\dot{v}}_{eb}^e = \underline{\ddot{r}}_{eb}^e = \underline{\ddot{r}}_{ib}^e$$

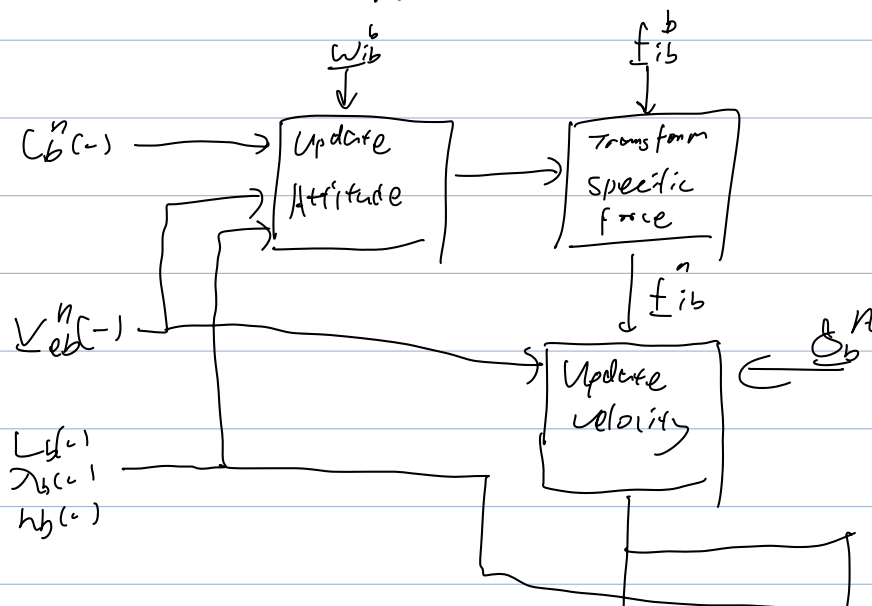
$$= -(\Omega_{ie}^e \Omega_{ie}^e + \dot{\Omega}_{ie}^e) \underline{r}_{ib}^e - 2 \Omega_{ie}^e \underline{\dot{r}}_{ib}^e + \underline{a}_{ib}^e$$

$$\Rightarrow \underline{\dot{v}}_{eb}^n = -\Omega_{en}^n \underline{v}_{eb}^n - (\Omega_{ie}^n \Omega_{ie}^n + 0) \underline{r}_{ib}^n - 2 \Omega_{ie}^n \underline{v}_{eb}^n + \underline{a}_{ib}^n$$

$$= -\Omega_{ie}^n \Omega_{ie}^n \underline{r}_{ib}^n - (\Omega_{en}^n + 2 \Omega_{ie}^n) \underline{v}_{eb}^n + \underline{a}_{ib}^n$$

$$\Rightarrow \underline{\dot{v}}_{eb}^n = C_b^n \underline{f}_{ib}^b + \underline{\delta}_b^n (L_b, h_b) - (\Omega_{en}^n + 2 \Omega_{ie}^n) \underline{v}_{eb}^n$$

need to be updated



$$\begin{array}{c}
 \downarrow \qquad \downarrow \downarrow \\
 \downarrow \qquad \boxed{\text{update positions}} \\
 \downarrow \qquad \downarrow \\
 \cup_{\phi_b(t)} \quad \cup_{\phi_b(t)} \cap_{\phi_b(t)} \quad h_b(t)
 \end{array}$$